

Week 2 easier

Week 2: Sampling Theory. Continuous-Time to Discrete-Time (C/D) conversion, D/C conversion, Nyquist theorem, aliasing, and quantization noise.

Week 2 easier

Can you map out the two 75-minute lectures for Week 2 covering Sampling, Aliasing, and C/D conversion?

Gemini Pro:

Here is the detailed breakdown for the two 75-minute lectures in Week 2.

This week bridges the physical world (continuous-time acoustic waves) and the digital domain (discrete-time arrays). It is vital to emphasize that any mistakes made during sampling (like aliasing) permanently destroy information, which will later ruin the performance of their machine learning models.

Lecture 3: Continuous-to-Discrete (C/D) Conversion & Aliasing

Objective: Mathematically model the sampling process using impulse trains, derive the frequency-domain spectrum of a sampled signal, and prove the Nyquist-Shannon Sampling Theorem.

Time Block	Topic & Mathematical Focus	Teaching Strategy & Audio Connection
00 - 15 min	The C/D Interface	Motivation: Microphones capture continuous-time voltage $x_c(t)$, but PyTorch only accepts discrete arrays $x[n]$. Define the relationship $x[n] = x_c(nT)$.
15 - 35 min	Ideal Impulse Train Sampling	Introduce the mathematical model. Multiply $x_c(t)$ by an impulse train $s(t) = \sum \delta(t - nT)$ to create the modulated signal $x_s(t)$.
35 - 55 min	Sampling in the Frequency Domain	Derive the spectrum of $x_s(t)$. Show that multiplication in time is convolution in frequency, resulting in infinite, scaled copies of the baseband spectrum separated by Ω_s .

Time Block	Topic & Mathematical Focus	Teaching Strategy & Audio Connection
55 - 75 min	Nyquist Theorem & Aliasing	Define the Nyquist rate mathematically: $\Omega_s > 2\Omega_N$. Draw what happens on the board when $\Omega_s < 2\Omega_N$ (the spectral copies overlap). <i>Audio ML connection:</i> Show how aliased high-frequency noise creates false patterns in a spectrogram.

Key Equations to Board

- **Impulse Train Sampling:**

$$x_s(t) = x_c(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x_c(nT) \delta(t - nT)$$

- **Spectrum of Sampled Signal:**

$$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

- **Continuous vs. Discrete Frequency:**

$$\omega = \Omega T$$

Lecture 4: Reconstruction, D/A, and Quantization Noise

Objective: Explain the mathematics of ideal reconstruction (D/C conversion), the realities of physical Digital-to-Analog hardware, and model the effects of bit-depth (quantization) on audio quality.

Time Block	Topic & Mathematical Focus	Teaching Strategy & Audio Connection
00 - 20 min	Ideal D/C Conversion	Explain how to isolate the baseband spectrum using an ideal continuous-time low-pass filter $H_r(j\Omega)$. Derive the time-domain interpolation formula using the sinc function.
20 - 40 min	Practical D/A: Zero-Order Hold	The math of ideal sinc interpolation is non-causal

Time Block	Topic & Mathematical Focus	Teaching Strategy & Audio Connection
		and infinitely long. Introduce the Zero-Order Hold (ZOH) circuit. Show how ZOH introduces a sinc-shaped envelope drop-off in the frequency domain.
40 - 55 min	Quantization & Bit-Depth	Time is discrete, but amplitude must also be discretized. Model quantization as an additive, wide-sense stationary noise source $e[n]$. Derive the 6.02 dB per bit rule.
55 - 75 min	Live Audio Demo	Take a clean vocal recording (24-bit, 48 kHz). Decimate it without an anti-aliasing filter to hear aliasing (sounds like robotic ringing). Then reduce it to 8-bit and 4-bit to hear quantization noise (sounds like loud white noise/fuzz).

Key Equations to Board

- **Ideal Bandlimited Interpolation:**

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin(\pi(t - nT)/T)}{\pi(t - nT)/T}$$

- **Quantization Noise Model:**

$$x_q[n] = x[n] + e[n]$$

- **Signal-to-Quantization-Noise Ratio (SQNR):**

$$\text{SQNR} \approx 6.02B + 1.76 \text{ dB}$$

(where B is the number of bits).

Instructor Tip: When discussing quantization, explicitly connect B (bit depth) to data types they will use in PyTorch. For instance, explain why raw audio is usually 16-bit integer (giving ~ 96 dB of dynamic range), but neural networks require casting these to 32-bit floats.

Prompt: Can you create slides and python code for these two 75 minute classes of a DSP and ML course for a combined senior and first year Electrical and Computer Engineering class? The DSP textbook is Discrete-Time Signal Processing by Oppenheim and Schaffer. Add some homework problems with solutions and a weekly quiz with solutions as well.

Chat GPT:

Done. I created the complete **Week 2 DSP + ML teaching package** for the two 75-minute classes.

Complete package

[Download the complete Week 2 package](#)

Individual files

- [Lecture 3 — Continuous-to-Discrete Conversion & Aliasing](#) — 22 slides with speaker notes and timing guidance.
- [Lecture 4 — Reconstruction, D/A, and Quantization Noise](#) — 24 slides with speaker notes and the live-demo sequence.
- [Week 2 Jupyter Notebook](#) — executed notebook with NumPy, SciPy, Matplotlib, PyTorch, spectrograms, and playable audio.
- [Standalone Python Code](#)
- [Homework + Weekly Quiz + Complete Solutions](#)

Audio files for the Lecture 4 demo

- [Clean 48 kHz reference](#)
- [Naive 48 → 12 kHz decimation with aliasing](#)
- [Anti-aliased 48 → 12 kHz decimation](#)
- [8-bit quantized audio](#)
- [4-bit quantized audio](#)

A few teaching choices I built in deliberately:

Lecture 3 treats aliasing as a non-uniqueness problem, not merely as vague “distortion.” Students see mathematically that a 2 kHz and 10 kHz cosine sampled at 12 kHz generate identical samples. That makes the argument that ML cannot magically recover lost information particularly strong.

The spectral-replica derivation goes all the way through

$$S(j\Omega) = \frac{2\pi}{T} \sum_k \delta(\Omega - k\Omega_s)$$

to

$$X_s(j\Omega) = \frac{1}{T} \sum_k X_c(j(\Omega - k\Omega_s)),$$

so the Nyquist condition follows geometrically rather than appearing as a rule to memorize.

Lecture 4 derives both sinc reconstruction and the ZOH response. The ZOH section shows why

$$H_{\text{ZOH}}(j\Omega) = T e^{-j\Omega T/2} \frac{\sin(\Omega T/2)}{\Omega T/2}$$

produces the familiar (-3.92) dB droop at Nyquist.

I also included a distinction that is often confusing in audio courses: the roughly **96 dB “6 dB/bit” figure for 16 bits versus the 98.08 dB full-scale-sine SQNR** obtained from $(6.02B+1.76)$.

The PyTorch material explicitly emphasizes that converting int16 PCM to float32 **changes the computational representation but does not recover quantization precision or undo aliasing.**

The homework contains **8 core problems plus a graduate stretch problem**, and the weekly quiz is a **20-point, approximately 15-minute assessment** with a grading guide. The notebook was executed successfully, both slide decks were rendered and checked for overflow, and the homework/quiz document was rendered page-by-page for layout verification.

Week 2, Lecture 3, LaTeX math

Here is the lecture content converted with properly formatted LaTeX for all the mathematical expressions, organized for clear readability:

This lecture addresses a fundamental question in signal acquisition:

When do samples contain enough information to uniquely represent the original analog signal?

The core mathematical progression flows as follows:

$x_c(t) \rightarrow$ impulse-train sampling $\rightarrow X_s(j\Omega) \rightarrow$ spectral replicas $\rightarrow$ Nyquist condition $\rightarrow$ aliasing if replicas overlap

Framing sampling as an **information-preservation problem** rather than merely taking measurements every T seconds highlights a critical machine learning consequence: if aliasing introduces false spectral structure before the data reaches a model, the model cannot distinguish that artificial structure from authentic acoustic features.

1. The C/D Interface: What Actually Changes When We Sample?

We begin with a continuous-time signal $x_c(t)$. For audio applications, this represents continuous physical phenomena such as microphone voltage.

Sampling every T seconds yields:

$$x[n] = x_c(nT)$$

where

$$f_s = \frac{1}{T}$$

Key variables:

- t : Continuous time variable (seconds)
- n : Discrete integer sample index
- T : Sampling period (seconds/sample)
- f_s : Sampling rate (samples/second)

Sampling discretizes **time**, whereas quantization discretizes **amplitude**.

Example 1

At $f_s = 48$ kHz, what physical time corresponds to sample $n = 2400$?

$$t = \frac{2400}{48000} = 0.05 \text{ s} \Rightarrow \boxed{50 \text{ ms}}$$

Example 2

Does ideal sampling change amplitude values?

- **Answer:** No. Ideal sampling merely evaluates the continuous-time function at discrete points. Amplitude restriction occurs later during quantization.

2. Continuous Frequency vs. Discrete Frequency

A continuous-time sinusoid is represented as:

$$x_c(t) = \cos(\Omega_0 t)$$

where Ω_0 is continuous angular frequency in rad/s. Sampling yields:

$$x[n] = \cos(\Omega_0 nT)$$

Defining the normalized discrete-time frequency ω_0 :

$$\omega_0 = \Omega_0 T$$

gives:

$$x[n] = \cos(\omega_0 n)$$

Here, ω carries units of rad/sample and is unique only modulo 2π . Because $\Omega = 2\pi f$, the relationship simplifies to:

$$\omega = 2\pi \frac{f}{f_s}$$

3. Normalized Frequency Mapping

At $f_s = 12$ kHz, the normalized discrete-time frequency corresponding to a 3 kHz sinusoid is:

$$\omega = 2\pi \frac{3}{12} = \frac{\pi}{2}$$

For a 15 kHz sinusoid at the same sample rate:

$$\omega = 2\pi \frac{15}{12} = \frac{5\pi}{2}$$

Evaluating modulo 2π :

$$\frac{5\pi}{2} \equiv \frac{\pi}{2}$$

Thus, 3 kHz and 15 kHz map to the exact same discrete-time complex exponential when sampled at 12 kHz.

4. The Impulse Train as an Analytical Tool

Ideal impulse-train sampling models physical sampling mathematically. Defining the periodic impulse train:

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

The sampled continuous-time signal $x_s(t)$ is:

$$x_s(t) = x_c(t)s(t) = x_c(t) \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

Applying the sifting property $x_c(t)\delta(t - nT) = x_c(nT)\delta(t - nT)$:

$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT) \delta(t - nT)$$

Each impulse location marks a sample instant, with the discrete sequence $x[n]$ embedded as the weights of the impulses.

5. Sample Weight Extraction

Consider $x_c(t) = \cos(2\pi \cdot 1000t)$ sampled at $f_s = 8000$ Hz ($T = \frac{1}{8000}$ s).

The impulse weights evaluate to:

$$x_c(nT) = \cos\left(2\pi \cdot 1000 \frac{n}{8000}\right) = \cos\left(\frac{\pi n}{4}\right)$$

Yielding the impulse-train representation:

$$x_s(t) = \sum_{n=-\infty}^{\infty} \cos\left(\frac{\pi n}{4}\right) \delta(t - nT)$$

The underlying discrete sequence is extracted directly from these weights: $x[n] = \cos\left(\frac{\pi n}{4}\right)$.

6. Frequency Representation of the Impulse Train

A periodic impulse train in the time domain transforms into an impulse train in the frequency domain. For $s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$, its Continuous-Time Fourier Transform (CTFT) is:

$$S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$$

where

$$\Omega_s = \frac{2\pi}{T} = 2\pi f_s$$

This relationship ($T \leftrightarrow \Omega_s = \frac{2\pi}{T}$) causes spectral replication in the frequency domain.

7. Derivation of Spectral Replicas

Because multiplication in time corresponds to convolution in frequency:

$$X_s(j\Omega) = \frac{1}{2\pi} X_c(j\Omega) * S(j\Omega)$$

Substituting $S(j\Omega)$:

$$X_s(j\Omega) = \frac{1}{2\pi} X_c(j\Omega) * \left[\frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s) \right]$$

Simplifying:

$$X_s(j\Omega) = \frac{1}{T} X_c(j\Omega) * \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$$

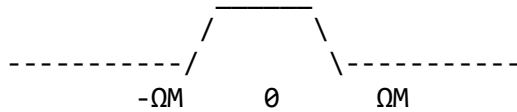
Since convolution with a shifted Dirac impulse shifts the spectrum $X_c(j\Omega) * \delta(\Omega - k\Omega_s) = X_c(j(\Omega - k\Omega_s))$:

$$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

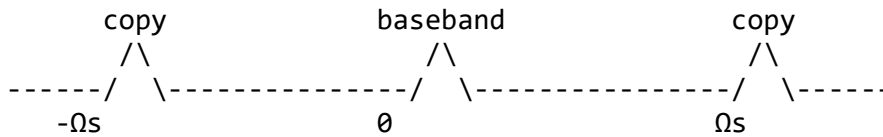
Sampling replicates the original continuous spectrum $X_c(j\Omega)$ at integer multiples of the sampling frequency Ω_s , scaled by $\frac{1}{T}$.

8. Spectral Overlay Visualization

Assuming $X_c(j\Omega)$ is bandlimited to Ω_M ($X_c(j\Omega) = 0$ for $|\Omega| > \Omega_M$):



Sampling produces infinitely many shifted copies centered at $0, \pm\Omega_s, \pm2\Omega_s, \dots$:



As long as neighboring spectral replicas do not overlap, the baseband signal remains intact.

9. Derivation of the Nyquist Condition

The baseband spectrum occupies the range $-\Omega_M \leq \Omega \leq \Omega_M$ (total width $2\Omega_M$).

The adjacent positive replica is centered at Ω_s , extending down to its left edge at $\Omega_s - \Omega_M$. To prevent spectral overlap:

$$\Omega_s - \Omega_M > \Omega_M \Rightarrow \Omega_s > 2\Omega_M$$

Expressed in Hz:

$$f_s > 2f_M$$

When this condition holds, an ideal low-pass filter with cutoff frequency $\Omega_c = \frac{\Omega_s}{2}$ isolates the original spectrum $X_c(j\Omega)$ perfectly.

10. Nyquist Frequency vs. Nyquist Rate

- **Nyquist Frequency (f_N):** A property of the **system**. Given sampling rate f_s :

$$f_N = \frac{f_s}{2}$$

(e.g., $f_s = 48 \text{ kHz} \Rightarrow f_N = 24 \text{ kHz}$)

- **Nyquist Rate:** A property of the **signal**. Given maximum signal frequency f_M :

$$\text{Nyquist Rate} = 2f_M$$

(e.g., $f_M = 18 \text{ kHz} \Rightarrow \text{Nyquist Rate} = 36 \text{ kHz}$)

11. Practical Guard Bands

While theoretical derivations state $f_s \geq 2f_M$, physical systems cannot construct ideal rectangular brick-wall filters. Practical anti-aliasing filters require a finite **transition band**:

$$f_M < \frac{f_s}{2}$$

Physical analog front-ends need operating margin between the maximum signal frequency and f_N .

12. Defining Aliasing

$$\text{Aliasing is non-uniqueness.}$$

When spectral replicas overlap, higher-frequency components fold into lower frequencies. Aliasing is non-invertible distortion; it creates deterministic, false sinusoidal components in the passband that are identical to real lower-frequency signals.

13. Indistinguishable Discrete-Time Sinusoids

Consider $f_s = 12$ kHz comparing $x_1(t) = \cos(2\pi \cdot 2000t)$ and $x_2(t) = \cos(2\pi \cdot 10000t)$.

Sampling $x_1(t)$:

$$x_1[n] = \cos\left(2\pi \frac{2000}{12000}n\right) = \cos\left(\frac{\pi n}{3}\right)$$

Sampling $x_2(t)$:

$$x_2[n] = \cos\left(2\pi \frac{10000}{12000}n\right) = \cos\left(\frac{5\pi n}{3}\right)$$

Using periodicity $\frac{5\pi}{3} = 2\pi - \frac{\pi}{3}$ and cosine symmetry:

$$\cos\left(\frac{5\pi n}{3}\right) = \cos\left(2\pi n - \frac{\pi n}{3}\right) = \cos\left(-\frac{\pi n}{3}\right) = \cos\left(\frac{\pi n}{3}\right)$$

$$\boxed{x_1[n] = x_2[n]}$$

Both continuous-time signals produce identical digital samples.

14. Irreversibility

Once sampled into $x[n] = \cos\left(\frac{\pi n}{3}\right)$, all historic context regarding whether the source frequency was 2 kHz or 10 kHz is lost. The sample sequence contains no metadata identifying its original continuous-time frequency.

15. The Folded Frequency Formula

For a continuous sinusoid of frequency f sampled at f_s , the observed alias frequency f_{alias} in the baseband $0 \leq f_{\text{alias}} \leq \frac{f_s}{2}$ satisfies:

$$f_{\text{alias}} = |f - kf_s| \quad \text{for } k \in \mathbb{Z}$$

For $f_s = 12$ kHz ($f_N = 6$ kHz):

- **7 kHz:** $|7 - 12| = 5$ kHz $\Rightarrow$ $\boxed{7 \text{ kHz} \rightarrow 5 \text{ kHz}}$
 - **9 kHz:** $|9 - 12| = 3$ kHz $\Rightarrow$ $\boxed{9 \text{ kHz} \rightarrow 3 \text{ kHz}}$
 - **10 kHz:** $|10 - 12| = 2$ kHz $\Rightarrow$ $\boxed{10 \text{ kHz} \rightarrow 2 \text{ kHz}}$
 - **13 kHz:** $|13 - 12| = 1$ kHz $\Rightarrow$ $\boxed{13 \text{ kHz} \rightarrow 1 \text{ kHz}}$
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16. Visualizing Spectral Folding

At $f_s = 12$ kHz, frequencies reflect symmetrically around the Nyquist boundary $f_N = 6$ kHz:

Input f (kHz)	Folded f_{alias} (kHz)
0	0
2	2
6	6
7	5
10	2
12	0

17. Normalized Frequency Perspective

Re-evaluating the 10 kHz input at $f_s = 12$ kHz in normalized frequency:

$$\omega = 2\pi \frac{10}{12} = \frac{5\pi}{3}$$

Modular reduction yields:

$$\frac{5\pi}{3} \equiv -\frac{\pi}{3} \pmod{2\pi}$$

Taking the real part for a cosine:

$$\cos\left(-\frac{\pi}{3}n\right) = \cos\left(\frac{\pi}{3}n\right)$$

Converting back to linear frequency in Hz:

$$f = \frac{\pi/3}{2\pi} \cdot 12000 = 2000 \text{ Hz}$$

18. Pre-ADC Filtering Requirements

If an analog input contains a desired 2 kHz component alongside an unwanted 10 kHz component, sampling at $f_s = 12$ kHz maps both to 2 kHz:

$$x[n] = A_1 \cos\left(2\pi \frac{2000}{f_s} n\right) + A_2 \cos\left(2\pi \frac{2000}{f_s} n\right) = (A_1 + A_2) \cos\left(2\pi \frac{2000}{f_s} n\right)$$

Because the terms combine linearly into a single 2 kHz discrete component, no digital filter can separate them post-sampling. The anti-aliasing filter must exist in the **analog domain prior to sampling**:

19. Digital Inability to Fix Aliasing

Digital post-processing cannot fix aliasing because violation of the Nyquist criterion represents a many-to-one mapping. Overlapping spectral content destroys linear independence between baseband and alias frequencies.

20. Downsampling / Decimation

Downsampling an already digitized signal requires pre-filtering. Converting an audio stream from 48 kHz to 12 kHz reduces the Nyquist frequency from 24 kHz to 6 kHz.

Low-pass filter first, decimate second.

- Direct slicing (`x[:4]`) without filtering causes high frequencies (6–24 kHz) to alias into the 0–6 kHz band.
 - Polyphase filtering (`scipy.signal.resample_poly`) suppresses content above 6 kHz prior to decimation.
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21. Mechanism of Naive Decimation Aliasing

Consider a 10 kHz component in a 48 kHz signal ($f_N = 24$ kHz). Selecting every 4th sample drops f_s to 12 kHz ($f_N' = 6$ kHz). The 10 kHz component now exceeds f_N' and folds:

$$|10 - 12| = 2 \text{ kHz} \Rightarrow 10 \text{ kHz} \rightarrow 2 \text{ kHz}$$

22. Proper Decimation Trade-offs

Proper decimation strips energy above the target Nyquist rate (6 kHz) using a digital low-pass filter before sample removal. Information above 6 kHz is lost by design, avoiding false low-frequency artifacts.

23. Downstream Machine Learning Vulnerabilities

If audio containing an un-filtered 10 kHz tone undergoes naive decimation, a 2 kHz line appears in the resulting spectrogram.

Strong energy at 2 kHz

A convolutional neural network (CNN) trained on these features will treat this line as valid 2 kHz acoustic data, potentially causing the model to learn hardware-dependent sampling artifacts rather than true physical class features.

24. Dataset Shortcuts

In machine learning, if Class A recordings originate from an ADC with poor anti-alias filtering while Class B recordings do not, a CNN may achieve 100% training accuracy strictly by classifying hardware artifacts. When deployed on standard hardware, performance collapses.

25. Application Problem 1

Scenario: A sensor signal is bandlimited to 18 kHz. Is $f_s = 32$ kHz sufficient?

- **Calculation:** Nyquist Rate = $2f_M = 2(18) = 36$ kHz.
 - **Evaluation:** $32 \text{ kHz} < 36 \text{ kHz}$.
 - **Verdict:** No. The replicas will overlap because $f_N = 16 \text{ kHz} < 18 \text{ kHz}$.
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26. Application Problem 2

Scenario: A 17 kHz tone is sampled at $f_s = 20$ kHz. What frequency appears in the output?

- **Calculation:** $f_N = 10$ kHz.
- **Alias Calculation:** $|17 - 20| = 3$ kHz.
- **Normalized Frequency Verification:**

$$\omega = 2\pi \frac{17}{20} = 1.7\pi \equiv -0.3\pi \pmod{2\pi}$$

$$f = \frac{0.3\pi}{2\pi} (20 \text{ kHz}) = \boxed{3 \text{ kHz}}$$

27. Application Problem 3

Scenario: Why can't a digital low-pass filter placed after the ADC remove aliasing?

Once sampled, the aliased component and legitimate in-band signal at the same frequency merge into a single sequence value. Information distinguishing their origins is destroyed.

28. Ultrasonic Interference

Even if useful acoustic signals top out at 20 kHz and $f_s = 48$ kHz ($f_N = 24$ kHz), out-of-band ultrasonic noise, radio frequency demodulation, or circuit harmonics above 24 kHz can still alias into the audible spectrum without an analog anti-aliasing filter.

29. Practical Bandlimiting

Theoretical signals satisfy $X_c(j\Omega) = 0$ for $|\Omega| > \Omega_M$. Real physical signals are rarely strictly bandlimited, requiring analog filtering to attenuate out-of-band energy below the dynamic range floor of the ADC.

30. Band Support Boundaries

Consider a continuous signal with spectral support in the range $[-4$ kHz, $+4$ kHz].

- **Case A** ($f_s = 12$ kHz): Replicas center at $0, \pm 12$ kHz.
 - Baseband: $[-4, +4]$ kHz
 - First replica: $[8, 16]$ kHz
 - *Result*: No spectral overlap.
 - **Case B** ($f_s = 6$ kHz): Replicas center at $0, \pm 6$ kHz.
 - Baseband: $[-4, +4]$ kHz
 - First replica: $[2, 10]$ kHz
 - *Result*: Overlap occurs in the $[2, 4]$ kHz region.
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31. Single Sinusoid Ambiguity

Aliasing is irreversible from the samples alone.

Without external prior assumptions regarding signal bandlimits, an infinite set of continuous-time frequencies $\{|f_{\text{alias}} \pm kf_s|\}$ maps to the exact same sample sequence.

32. Physical ADC Architecture

A complete continuous-to-digital hardware chain contains distinct functional blocks:

Analog Input $\rightarrow$ Anti-Alias Filter $\rightarrow$ Sample-and-Hold $\rightarrow$ Quantizer $\rightarrow$ Digital Code

33. Aliasing vs. Quantization Noise

- **Aliasing:** Caused by under-sampling in time ($f_s < 2f_M$). Rotates high-frequency energy into false lower frequencies.

Aliasing changes frequency identity.

- **Quantization:** Caused by finite bit-depth resolution in amplitude. Modeled as additive noise $e[n] = x_q[n] - x[n]$.

Quantization introduces amplitude error.

34. Checkpoint Questions

1. **A 5 kHz sinusoid is sampled at 8 kHz. What frequency appears digitally?**
 - Calculation: $|5 - 8| = 3$ kHz.
 - **Answer:** 3 kHz.
 1. **A signal is bandlimited to 6 kHz. Which rate satisfies Nyquist with margin?**
 - Calculation: $2f_M = 12$ kHz. Rate must strictly exceed 12 kHz.
 - **Answer:** 16 kHz.
 1. **Which component must appear first in an ADC front-end?**
 - **Answer:** Analog anti-aliasing filter.
 1. **Why is aliasing problematic for ML models?**
 - **Answer:** It introduces false, structured spectral features into the input data.
 1. **At $f_s = 16$ kHz, which two frequencies produce identical discrete cosines?**
 - Calculation: $|16 - 13| = 3$ kHz.
 - **Answer:** 3 kHz and 13 kHz.
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35. Core Concepts

- **Spectral Replication Origin:** Time-domain multiplication with an impulse train corresponds to frequency-domain convolution with an impulse train.
 - **Factor of 2 in Nyquist:** A real bandlimited spectrum extending from $-f_M$ to $+f_M$ occupies a bandwidth of $2f_M$.
 - **Anti-Alias Filtering Location:** Must occur in the analog domain to remove energy above f_N before sampling makes it indistinguishable from lower-frequency content.
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36. Information-Theoretic Summary

No overlap $\Rightarrow$ unique reconstruction; overlap $\Rightarrow$ aliasing and non-uniqueness

If $f_s > 2f_M$, spectral replicas remain isolated, guaranteeing unique signal recovery. If $f_s < 2f_M$, spectral overlap destroys uniqueness.

37. Final Conceptual Mapping

$$x_c(t) \xrightarrow{\text{sample every } T} x[n]$$

$$x_c(t) \times \sum_n \delta(t - nT)$$

$$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

Spectral replicas

$$f_s > 2f_M \Rightarrow \text{Replicas separate}$$

$$f_s < 2f_M \Rightarrow \text{Replicas overlap} \Rightarrow \text{Aliasing}$$

Once false structure enters the samples, downstream models treat it as valid data.

That is the larger reason this lecture belongs early in a DSP + ML course: students learn that **the data acquisition chain is part of the machine-learning system**, not merely something that happened before the “interesting” ML began.

Week 2, Lecture 4, LaTeX math

Here is the lecture content converted with properly formatted LaTeX for all the mathematical expressions, organized for clear readability:

This lecture covers signal reconstruction, continuous-to-discrete amplitude mapping, and the theoretical and practical foundations of quantization.

When do samples faithfully represent the original analog signal?

Lecture 4 addresses two key follow-up questions:

If the samples are valid, how do we turn them back into a continuous waveform?

What happens when amplitudes can only be represented with a finite number of bits?

This presents two distinct analytical paths:

samples → ideal reconstruction → practical DAC/ZOH

continuous amplitude → quantization → error/noise → SQNR

Sampling error (aliasing) shifts energy to incorrect frequencies and creates non-uniqueness, whereas quantization maps continuous amplitudes onto a finite set of discrete levels.

1. Distinguishing the Two Kinds of Discretization

- **Sampling Discretizes Time:** Starting with continuous-time signal $x_c(t)$, sampling yields discrete-time sequence:

$$x[n] = x_c(nT)$$

In an ideal mathematical model, amplitudes $x[n]$ remain unconstrained continuous real values.

- **Quantization Discretizes Amplitude:** Quantization maps continuous amplitude values to discrete levels.

sampling: continuous time → discrete time

quantization: continuous amplitude → discrete amplitude levels

2. Ideal Reconstruction: Signal Recovery

Ideal sampling creates spectral replicas:

$$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

When the Nyquist criterion is satisfied, adjacent copies do not overlap. Reconstructing the original signal requires isolating the baseband spectrum centered at $\Omega = 0$ while rejecting all other copies using an ideal reconstruction low-pass filter:

$$H_r(j\Omega) = \begin{cases} T, & |\Omega| < \Omega_c \\ 0, & \text{otherwise} \end{cases}$$

3. Reconstruction Filter Gain T

Inside the baseband region, the sampled spectrum is:

$$X_s(j\Omega) = \frac{1}{T} X_c(j\Omega)$$

Applying a unit-gain filter leaves an amplitude scaling of $\frac{1}{T}X_c(j\Omega)$. Setting filter gain $H_r(j\Omega) = T$ in the passband balances the scale factor:

$$T \left(\frac{1}{T} X_c(j\Omega) \right) = X_c(j\Omega)$$

The filter does not construct missing signal information; under bandlimited conditions, the samples contain sufficient information to uniquely specify the continuous waveform.

4. Mathematical Origin of Sinc Interpolation

An ideal rectangular brick-wall filter in the frequency domain corresponds to a sinc function in the time domain:

$$h_r(t) = \text{sinc}\left(\frac{t}{T}\right)$$

Convolving the impulse train with $h_r(t)$ yields the ideal continuous-time reconstruction formula:

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] \text{sinc}\left(\frac{t - nT}{T}\right)$$

where normalized sinc is defined as:

$$\text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}$$

Each sample weight scales a centered, shifted sinc kernel.

5. Physical Meaning of Sinc Interpolation

Every sample $x[n]$ scales a sinc function centered at $t = nT$. The reconstructed signal is the infinite sum of all shifted kernels across time:

$$x_r(t) = \dots + x[-1] \text{sinc}\left(\frac{t + T}{T}\right) + x[0] \text{sinc}\left(\frac{t}{T}\right) + x[1] \text{sinc}\left(\frac{t - T}{T}\right) + \dots$$

Because the sinc function has infinite support, every sample influences intermediate signal values across all time.

6. Sample Point Exact Intersections

The normalized sinc function satisfies:

$$\text{sinc}(0) = 1 \quad \text{and} \quad \text{sinc}(k) = 0 \quad \forall k \in \mathbb{Z}, k \neq 0$$

Evaluating reconstruction at sample instant $t = mT$:

$$x_r(mT) = \sum_{n=-\infty}^{\infty} x[n] \text{sinc}(m - n)$$

All terms evaluate to zero except where $n = m$, yielding:

$$x_r(mT) = x[m]$$

At $t = 3T$, the contribution from the sample at $n = 1$ uses argument $3 - 1 = 2$:

$$\text{sinc}(2) = 0 \Rightarrow \boxed{0}$$

7. Uniqueness Between Sample Points

For arbitrary unconstrained functions, infinitely many curves pass through a set of discrete points. Under the assumption that the original signal is strictly bandlimited below $\frac{f_s}{2}$, only **one unique continuous signal** can generate those exact sample values.

8. Physical DAC Constraints

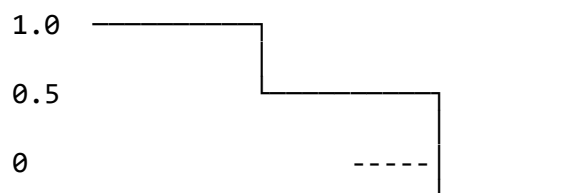
Ideal sinc interpolation cannot be implemented directly in physical hardware due to:

- Infinite time duration (infinite support)
- Non-causality ($h_r(t) \neq 0$ for $t < 0$)
- Unachievable ideal rectangular frequency response
- Incomplete bandlimiting in physical inputs

Hardware uses causal continuous-time approximations, such as the Zero-Order Hold (ZOH).

9. Zero-Order Hold (ZOH) Functionality

A ZOH converter maintains a constant voltage output equal to the most recent sample value until the subsequent clock edge:



-0.2
0 T 2T

The resulting piecewise-constant staircase is a continuous-time analog signal containing high-frequency spectral images and low-pass magnitude droop.

10. ZOH as Pulse Convolution

Impulse-train representation:

$$x_s(t) = \sum_{n=-\infty}^{\infty} x[n] \delta(t - nT)$$

Defining the rectangular pulse $p(t)$:

$$p(t) = \begin{cases} 1, & 0 \leq t < T \\ 0, & \text{otherwise} \end{cases}$$

The ZOH output equals the convolution of impulses with the pulse shape:

$$x_{\text{ZOH}}(t) = x_s(t) * p(t)$$

impulse * rectangle = shifted rectangle

11. ZOH Frequency Response Derivation

Taking the Fourier transform of pulse $p(t) = u(t) - u(t - T)$:

$$H_{\text{ZOH}}(j\Omega) = \int_0^T e^{-j\Omega t} dt = \frac{1 - e^{-j\Omega T}}{j\Omega}$$

Factoring symmetrically:

$$1 - e^{-j\Omega T} = e^{-j\Omega T/2} (e^{j\Omega T/2} - e^{-j\Omega T/2}) = 2je^{-j\Omega T/2} \sin\left(\frac{\Omega T}{2}\right)$$

Substituting into the transfer function:

$$H_{\text{ZOH}}(j\Omega) = T e^{-j\Omega T/2} \frac{\sin(\Omega T/2)}{\Omega T/2}$$

$H_{\text{ZOH}}(j\Omega) = T e^{-j\Omega T/2} \text{sinc}\left(\frac{\Omega T}{2\pi}\right)$

This response consists of a linear-phase delay term combined with a sinc magnitude envelope.

12. Linear-Phase Delay Factor

The exponential factor $e^{-j\Omega T/2}$ represents a fixed time delay of half the sample period:

$$t_0 = \frac{T}{2}$$

This delay arises because the holding pulse extends forward in time from $t = 0$ to $t = T$ rather than being centered at $t = 0$.

13. ZOH Magnitude Response & Attenuation

Ignoring constant scale factor T :

$$\left| \frac{H_{\text{ZOH}}(j\Omega)}{T} \right| = \left| \frac{\sin(\Omega T/2)}{\Omega T/2} \right|$$

- Near DC ($\Omega \approx 0$): $\frac{\sin x}{x} \approx 1$
- Higher frequencies suffer increasing attenuation up to the first spectral zero at $\Omega = \frac{2\pi}{T}$

ZOH attenuates high frequencies more than low frequencies.

14. Quantifying Nyquist Droop

At the digital Nyquist limit ($\Omega T = \pi$):

$$\frac{|H|}{T} = \frac{\sin(\pi/2)}{\pi/2} = \frac{2}{\pi} \approx 0.6366$$

Expressed in decibels:

$$20\log_{10}(0.6366) \approx -3.92 \text{ dB}$$

This -3.92 dB attenuation near Nyquist is deterministic frequency shaping inherent to holding continuous samples, not random noise.

15. Role of the Analog Reconstruction Filter

The ZOH envelope attenuates high-frequency spectral replicas centered at $f_s, 2f_s, \dots$ but does not eliminate them entirely:

$x[n] \rightarrow \text{DAC/ZOH} \rightarrow \text{Analog Reconstruction LPF} \rightarrow x_r(t)$

The post-DAC analog low-pass filter suppresses remaining high-frequency images.

16. Oversampling Advantage

- **Base Rate Sampling** ($f_s = 44.1$ kHz): Passband ends at 20 kHz and Nyquist is 22.05 kHz, requiring steep analog filter roll-off in a 2.05 kHz transition band.
 - **Oversampled Conversion** ($f_s = 176.4$ kHz): First spectral images are pushed up to 176.4 kHz, allowing a gradual analog filter transition band.
-

17. Quantization Dynamics

Quantization converts continuous amplitude values into B -bit numerical codes:

$$L = 2^B$$

Resolution (B)	Code Levels ($L = 2^B$)
4 bits	16
8 bits	256
16 bits	65,536
24 bits	16,777,216

Inputs exceeding the input voltage range $[-A, A)$ cause saturation clipping.

18. Quantizer Step Size Formula

For dynamic range $[-A, A)$ with total width $2A$ split across 2^B levels:

$$\Delta \approx \frac{2A}{2^B}$$

- For $A = 1, B = 8$: $\Delta \approx \frac{2}{256} = 0.0078125$
 - For $A = 1, B = 4$: $\Delta \approx \frac{2}{16} = 0.125$
-

19. Additive Quantization Error Model

Quantization error $e[n]$ is defined as:

$$e[n] = x_q[n] - x[n] \Rightarrow x_q[n] = x[n] + e[n]$$

For round-to-nearest quantizers, error bounds are:

$$-\frac{\Delta}{2} \leq e[n] < \frac{\Delta}{2}$$

20. Statistical Noise Model Assumptions

Quantization is deterministic, but error sequence $e[n]$ can be modeled as zero-mean uniform white noise:

$$e[n] \sim \text{Uniform}\left(-\frac{\Delta}{2}, \frac{\Delta}{2}\right)$$

This model applies when signal amplitude spans many quantization steps, possesses complex phase structure, and quantization resolution is sufficiently high.

21. Derivation of Error Variance σ_e^2

For uniformly distributed random variable $e \sim U\left(-\frac{\Delta}{2}, \frac{\Delta}{2}\right)$ with $p(e) = \frac{1}{\Delta}$:

$$\begin{aligned}\sigma_e^2 &= \mathbb{E}[e^2] = \int_{-\Delta/2}^{\Delta/2} e^2 \left(\frac{1}{\Delta}\right) de \\ \sigma_e^2 &= \frac{1}{\Delta} \left[\frac{e^3}{3} \right]_{-\Delta/2}^{\Delta/2} = \frac{1}{\Delta} \left(\frac{\Delta^3}{24} - \left(-\frac{\Delta^3}{24} \right) \right) \\ \sigma_e^2 &= \frac{\Delta^2}{12}\end{aligned}$$

22. Derivation of 6 dB/Bit Rule

Incrementing bit depth by 1 bit doubles available levels:

$$2^B \rightarrow 2^{B+1} \Rightarrow \Delta \rightarrow \frac{\Delta}{2}$$

Error power scales quadratically with step size:

$$\sigma_e^2 \propto \Delta^2 \Rightarrow \sigma_{e,\text{new}}^2 = \frac{(\Delta/2)^2}{12} = \frac{1}{4} \sigma_e^2$$

In decibels:

$$10 \log_{10}(4) \approx 6.02 \text{ dB}$$

one additional bit $\approx$ 6.02 dB improvement

23. Full-Scale Sinusoid SQNR Derivation

For full-scale sinusoid $x[n] = A\sin(\omega n)$:

- RMS Amplitude: $\frac{A}{\sqrt{2}}$
- Signal Power: $P_x = \frac{A^2}{2}$
- Step Size: $\Delta = \frac{2A}{2^B}$
- Noise Power: $P_e = \frac{\Delta^2}{12} = \frac{1}{12} \frac{4A^2}{2^{2B}} = \frac{A^2}{3 \cdot 2^{2B}}$

Calculating Signal-to-Quantization-Noise Ratio:

$$\text{SQNR} = \frac{P_x}{P_e} = \frac{A^2/2}{A^2/(3 \cdot 2^{2B})} = \frac{3}{2} \cdot 2^{2B}$$

Converting to decibels:

$$\text{SQNR}_{\text{dB}} = 10\log_{10}\left(\frac{3}{2}\right) + 20B\log_{10}(2)$$

$$\boxed{\text{SQNR} \approx 6.02B + 1.76 \text{ dB}}$$

24. Standard SQNR Reference Values

- **8-bit:** $6.02(8) + 1.76 = 49.92 \text{ dB} \approx \boxed{50 \text{ dB}}$
 - **12-bit:** $6.02(12) + 1.76 = 74.00 \text{ dB} \approx \boxed{74 \text{ dB}}$
 - **16-bit:** $6.02(16) + 1.76 = 98.08 \text{ dB} \approx \boxed{98.1 \text{ dB}}$
-

25. 96 dB vs. 98.1 dB Dynamic Range Distinction

- $6.02 \times 16 = 96.32 \text{ dB}$ estimates raw 16-bit dynamic range.
 - $6.02(16) + 1.76 = 98.08 \text{ dB}$ calculates ideal SQNR specifically for a full-scale sinusoid.
-

26. Signal Power Dependencies

The 1.76 dB constant assumes a full-scale sinusoid. If signal amplitude drops to $0.1A$ (-20 dB input level) while step size Δ remains fixed, actual SQNR drops by 20 dB.

27. Quantization vs. Overload Clipping

- **Quantization Error:** Bounded round-off within $[-\frac{\Delta}{2}, \frac{\Delta}{2})$.

- **Clipping:** Non-linear waveform truncation occurring when $|x(t)| > A$. Creates severe harmonic distortion.
-

28. Coarse Quantization Artifacts (4-bit Audio)

At low resolutions ($B = 4, 16$ levels), error terms correlate strongly with signal trajectories, generating coarse, harmonic distortion rather than white additive noise.

29. Low-Amplitude Periodic Error

Low-amplitude sinusoids crossing only 2–3 quantization levels yield periodic error signals. Periodic errors introduce distinct harmonic spectral lines instead of uniform noise floors.

30. Function of Dither

Dither adds small-amplitude random noise (e.g., triangular probability density function with amplitude $\pm\Delta$) prior to quantization. Dither decorrelates quantization error from the input signal, converting structured harmonic distortion into stationary, non-objectionable white noise.

structured distortion $\rightarrow$ randomized noise floor

31. Machine Learning Sensitivity to Artifacts

Models trained on coarsely quantized inputs without dither may learn false, structured amplitude artifacts as valid class features.

32. Numerical Type Casting (int16 to float32)

Converting 16-bit PCM integer samples to floating-point tensors changes computational precision, not underlying acquisition measurement resolution.

```
x = pcm.astype(np.float32) / 32768.0 # Scales numeric range to [-1.0, 1.0)
```

33. Information Invariance Under Type Conversion

Casting an int16 value to float32 does not recover amplitude detail removed during quantization.

more bits in container $\neq$ more measurement detail

34. Non-Reversibility of Hardware Distortion

Type casting to float64 cannot restore information lost to aliasing or quantization.

35. Precision Conceptual Check

Loading 16-bit audio into PyTorch as a torch.float32 tensor updates the arithmetic container format used for gradient updates and operations, leaving physical measurement resolution fixed at 16 bits.

36. Audio Demonstration Progression

1. **Clean 48 kHz:** Baseline reference.
 2. **Naive 48 kHz → 12 kHz:** Severe aliasing fold-over (false frequencies).
 3. **Filtered 48 kHz → 12 kHz:** Bandwidth reduced above 6 kHz without aliasing distortion.
 4. **8-bit Quantization:** Higher noise floor, speech remains intelligible.
 5. **4-bit Quantization:** Heavy signal-correlated granular distortion.
-

37. Qualitative Distinction: Aliasing vs. Quantization

- **Aliasing:** Maps spectral energy to incorrect baseband locations, causing irreversible frequency overlap.
 - **Quantization:** Perturbs sample amplitude values, adding noise or harmonic distortion.
-

38. Structural Comparison: ZOH Droop vs. Quantization Noise

- **ZOH Droop:** Deterministic continuous-time linear magnitude attenuation (sinc shaped).
 - **Quantization Noise:** Amplitude error statistical approximation ($\frac{4^2}{12}$ variance).
-

39. Core Conceptual Answers

- **Ideal Reconstruction Sinc Basis:** The ideal low-pass reconstruction filter has a rectangular spectrum in frequency, corresponding to a $\text{sinc}(t)$ impulse response in time.
- **ZOH Frequency Impact:** Introduces a sinc-shaped attenuation envelope (-3.92 dB at Nyquist) and a $T/2$ linear-phase delay.
- **12-bit Full-Scale Sine SQNR:**

$$\text{SQNR} = 6.02(12) + 1.76 = \boxed{74.0 \text{ dB}}$$

40. Verification Questions

1. **Sinc Reconstruction Contribution:** Each sample weight launches an infinitely long shifted sinc function.
 2. **Zero-Order Hold Core Function:** Holds sample amplitudes constant across sampling interval T .
 3. **Adding One Bit:** Step size Δ halves, noise power drops by a factor of 4, improving SQNR by ≈ 6 dB.
 4. **int16 to float32 Conversion:** Changes internal compute precision without increasing acquisition information.
 5. **Unmodeled Behavior:** Overload clipping falls outside the standard $\frac{\Delta^2}{12}$ uniform noise model.
-

41. Level-Based Analysis Summary

- **Undergraduate/Senior Formulations:**

$$\Delta \approx \frac{2A}{2^B}, \quad \sigma_e^2 = \frac{\Delta^2}{12}, \quad \text{SQNR} \approx 6.02B + 1.76 \text{ dB}$$

- **Graduate Formulations:** Integration of $e^2 p(e)$, bounds of uniform noise models, signal-error correlation dynamics, decorrelation via dither, and ZOH Fourier transform derivations.
-

42. End-to-End Acquisition & Reconstruction Chain

1. **Anti-Alias Filter:** $x_c(t) \rightarrow$ Analog LPF (suppresses frequencies above f_N).
 2. **Sampling:** $x[n] = x_c(nT)$ (discretizes time).
 3. **Quantization:** $x_q[n] = x[n] + e[n]$ (discretizes amplitude).
 4. **Type Casting:** int16 $\rightarrow$ float32 (prepares storage container for computation).
 5. **D/A Conversion:** ZOH produces staircase waveform $x_{\text{ZOH}}(t) = x_s(t) * p(t)$.
 6. **Reconstruction Filter:** Suppresses spectral images around kf_s .
-

43. Hardware-Software Pipeline Principles

Sampling determines whether frequency information survives.

Quantization determines how accurately amplitude survives.

Reconstruction determines how digital values map back to physical waveforms.

That gives students a much stronger conceptual basis than treating “ADC,” “DAC,” “bit depth,” and “sample rate” as isolated engineering vocabulary.
